AGENDA-SETTING AND TAX REFERENDA: IMPLICATIONS FOR REGRESSION DISCONTINUITY IDENTIFICATION STRATEGY USING ELECTION OUTCOMES

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Election outcomes frequently have been used as a source of identification in regression discontinuity (RD) designs by political scientists and economists. This paper tests the validity of using this type of RD identification strategy in county, municipality, township, and school district tax referendum elections where the taxing authority has some discretion over the referendum structure and timing. Using optimal bandwidth specifications, we find discontinuities at the majority vote threshold for numerous referendum and election characteristics that correspond with referendum aspects over which taxing authorities have discretion, and we do not find discontinuities in characteristics over which taxing authorities have less discretion. In addition, when considering the distribution of election closeness, we find a change in the probability across the majority vote threshold for these types of referenda. These findings bring into question the validity of the RD identification strategy when the taxing authority has agenda-setting ability and the bandwidth selection is such that the estimated discontinuity uses election outcomes that are not extremely close to the cutoff.

Keywords: regression discontinuity, local referenda, housing prices

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I. INTRODUCTION

Over the past several decades, economics research has been largely guided through a credibility revolution based on the rise of numerous causal inference techniques such as difference-in-differences and regression discontinuity

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(RD). The importance of this credibility revolution to the field of economics was made evident by the 2021 Nobel Prize in economics, which was awarded to several of those who sparked this movement toward causal estimation in economics. Since the late 1990s, RD has become an important tool in this credibility revolution (Cunningham, 2021).[1] Due to the recent rise in accessible data, election outcomes are frequently used as a source of identification in RD designs by economists. Elections often have the properties needed for an effective RD design, namely an explicit cutoff (e.g., majority vote threshold), a continuous running variable (e.g., vote share), and an inability of interested parties to precisely manipulate vote share. These attributes have resulted in a large body of research that uses an RD identification strategy on election outcomes to address a diverse set of economic issues including how local tax revenue, and thereby expenditures on local public goods, affects home prices. While Cellini, Ferreira, and Rothstein (2010) is one of the earliest papers that use tax referendum elections in an RD framework to estimate a causal impact on home prices, there is a substantial literature that followed using tax election–based RD design.[2]

The RD design, in theory, can avoid the challenges associated with adequately controlling for differences in communities that experience changes in their local tax revenue. The RD design compares home prices in communities where the tax referendum barely failed to those in communities that barely passed the tax referendum and estimates a discontinuity at the majority vote threshold based on this comparison. Conditional on using election outcomes close to the majority vote threshold, the RD identification strategy would provide causal estimates as long as potentially relevant variables, correlated with both election closeness and home prices, are continuous at this threshold/cutoff. However, if elections not very close to the majority vote threshold are used to estimate the discontinuity, then having relevant variables continuous at this threshold is not sufficient to allow a causal interpretation of the discontinuity estimate. In their 2010 *Journal of Economic Literature* article providing an overview of the issues of manipulation and bandwidth selection (i.e., choice of how close to the cutoff observations should be to be included in estimation) when considering the validity of the RD identification strategy, Lee and Lemieux discuss

[1] The RD method has been applied in many settings across economics to address important questions surrounding the impacts of health insurance, environmental regulations, education, political economy, and unionization. For examples of papers using RD to examine the impacts of health insurance, see Medicare-related work by Card, Dobkin, and Maestas (2008, 2009). RD papers related to environmental regulations primarily relate to air quality (e.g., Chay and Greenstone, 2005; Davis, 2008; Auffhammer and Kellogg, 2011; Chen and Whalley, 2012; Gallego, Montero, and Salas, 2013). Examples of RD designs have also been found in literature relating to education (e.g., Angrist and Lavy, 1999; Black, 1999; van der Klaauw, 2002; Chay, McEwan, and Urquiola, 2005; Solis, 2017), political economy (e.g., Lee, Moretti, and Butler, 2004; Lee, 2008), and unionization (e.g., DiNardo and Lee, 2004; Sojourner and Yang, 2022).

[2] See Isen (2014), Hong and Zimmer (2016), Martorell, Stange, and McFarlin (2016), Brasington (2017), Dronyk-Trosper (2017), Kogan, Lavertu, and Peskowitz (2017), Abott et al. (2020), Brasington and Zachariadis (2021), Enami, Alm, and Aranda (2021), Wei (2021), and Baron (2022).

the trade-offs associated with specifying a narrow bandwidth and using only observations close to the threshold/cutoff (imprecise estimates) and a wider bandwidth making "the comparisons on both sides of the cutoff less credible, as we are no longer comparing observations just to the left and right of the cutoff" (p. 308).[3] Interestingly, Lee and Lemieux consider this trade-off using vote share in US House of Representatives elections to estimate the causal effect of incumbency on election outcomes and demonstrate that using a bandwidth higher than 0.05 (i.e., elections where 45–55 percent vote for the Democratic candidate) is problematic based on two alternative specification tests.

Using 14,519 election results from municipality, township, school district, and county tax referenda from Ohio, this paper provides evidence that the election-based RD design has limitations when the discontinuity estimates are based on election outcomes that are not extremely close to the majority vote threshold/cutoff. While the agenda-setting abilities of the city councils, township trustees, school boards, and county commissioners are unlikely to enable *precise* manipulation of election outcomes at the majority vote threshold, these agenda setters often have discretion over multiple referenda characteristics, and this discretion enables them to manipulate election outcomes to a greater extent than in other types of elections, such as candidate-based elections.[4] Agenda setting in the context of local tax elections can take many forms, including the amount of the proposed tax, the timing of when a referendum is on the ballot, the type of tax, and what the tax will be spent on. Because the agenda setters are unlikely to have enough information on how referendum characteristics map into voter preferences to precisely manipulate election outcomes at the majority vote threshold, we would not expect biased estimates of housing price discontinuities if only elections where 49–51 percent vote yes (i.e., a bandwidth size of .01) are used in the estimation. However, it may be the case that this agenda setting changes the composition and distribution of outcomes in elections used in the discontinuity estimation when wider bandwidths are selected. The bandwidth sizes used in this paper, which are larger than 0.01 but similar to those in the existing literature and based on the optimal bandwidth derived in Calonico et al. (2017), provide empirical results that suggest the agenda setters' manipulation results in biased estimates of these housing price discontinuities.

The major advantage of this paper over the existing literature using an RD identification strategy on tax election outcomes is our detailed information on referendum and election characteristics. These referendum and election characteristics include the amount of the proposed tax, the type of tax, the timing of when a referendum is on ballot, and what the tax will be spent on. While the agenda-setting abilities along

3 Lee and Lemieux (2010) also discuss the issue arising from a wide bandwidth "where the mean value of Y over the whole bin is not representative of the mean value of Y at the boundaries of the bin" (p. 309).

4 There is extensive research using the RD framework on elections not involving tax referenda to obtain causal estimates for a variety of economic outcomes. See DiNardo and Lee (2004), Leigh (2008), Pettersson-Lidbom (2008), Ferreira and Gyourko (2009), Gerber and Hopkins (2011), Dell (2015), and Kogan, Lavertu, and Peskowitz (2021).

these multiple dimensions may allow more precise manipulation than in other types of elections, this paper focuses on whether the precision of this manipulation would result in potentially biased discontinuity estimates when election outcomes not extremely close to the cutoff are used in the estimation. Eggers et al. (2015) consider the validity of using an election-based RD design and emphasize that researchers should "consider theoretical mechanisms that could produce sorting around the discontinuity" and, when evaluating discontinuities in pretreatment covariates, focus "on the presence or absence of substantively large imbalances in characteristics that might be related to the mechanisms that could produce sorting."[5] In our setting, because a taxing authority often has discretion on a limited set of referendum characteristics, it is more likely that cutoff discontinuities would be estimated for these characteristics, due to these compositional and distributional changes, than for characteristics that the taxing authority cannot manipulate.

Like many states, Ohio imposes constraints on the referendum structure and timing that vary across taxing authorities and types of referenda. Overall, we estimate statistically significant discontinuities at the majority vote threshold (i.e., closeness cutoff) for referendum and election characteristics over which the taxing authority has discretion and, just as important, we do not find statistically significant discontinuities in referendum and election characteristics over which the taxing authority has less discretion. The bandwidth selections associated with these estimates are based on those derived in Calonico et al. (2017) and range from 0.10 (elections where 40–60 percent vote yes) to 0.23 (elections where 27–73 percent vote yes).[6] When considering the distribution of election closeness, after excluding referenda over which we expect the taxing authority to have less agenda-setting ability (i.e., referenda involving a renewal tax and current general expenditures), we find a statistically significant discontinuity in probability at the closeness cutoff. Finally, we assess the implications of these covariate imbalances for housing price analyses commonly conducted in this literature. Given these covariate imbalances, we show that the estimated discontinuities in home sale prices at the cutoff are sensitive to whether we control for referendum and election characteristics. These results suggest that election contests where authorities have a high degree of discretion on how and when the referendum appears on the ballot may be problematic for researchers using somewhat wide bandwidths in an RD identification strategy to identify causal effects of referenda passage on outcomes such as local tax revenues, expenditures,

[5] Eggers et al. (2015) consider the validity of the RD design in candidate-based elections. Eggers et al. (2015, p. 260) conclude that "the RD design [based on candidate elections] is a fundamentally sound and widely applicable approach to learning about the effect of election results on a variety of political and economic outcomes." However, Eggers et al. (2015) also note that an RD approach using candidate elections is only applicable for elections where candidates do not have viable mechanisms or actions by which to influence election outcome or are unable to accurately predict how these restricted actions will affect the election outcome.

[6] While the point estimates do not change appreciably when we decrease bandwidths to 0.05, the standard errors do increase as fewer observations are used when estimating the discontinuity.

and housing prices. While the agenda-setting ability of local tax authorities should be a concern for researchers interested in using an RD identification strategy on election outcomes from tax referenda, it may be less problematic in election contests where the agenda-setting ability is less pronounced and where the selected bandwidths are narrow.

II. CONCEPTUAL FRAMEWORK AND BACKGROUND ON OHIO TAX REFERENDA

A vast literature on the determinants of referendum election outcomes has shown that these outcomes clearly depend on the specifics of the referendum, such as the tax rate specified, how the revenue is raised (property, income, or sales tax), the restrictions imposed on how tax revenue is to be spent, whether or not the tax revenue is an extension of an existing tax (a renewal or a new tax), and when it appears on the ballot.[7] These referendum specifics will depend on the incentives of and information available to the agenda setter, as well as the discretion of the agenda setter along these different dimensions.

When considering incentives, Romer and Rosenthal (1978, 1979) model the objective function by assuming the agenda setter is a Niskanen (1975) bureaucrat who seeks to maximize the expected budget. Based on the desire to maximize the expected budget, the Niskanen bureaucrat would care about the tax rate specified in the referendum and the probability the referendum passes. The objective functions of city councils, township trustees, school boards, and county commissioners are also likely to depend on tax rate and whether the referendum passes, along with other aspects of the referendum such as the type of expenditures identified in the referendum, the "fungibility" of the tax revenue, whether it is a renewal or new tax, and how the burden of the tax varies across potential voters (especially based on whether it is a property, income, or sales tax). The taxing authorities understand that the probability of a referendum passing depends on the set of individuals who vote and that the composition of voters will depend not only on the referendum characteristics but also on when and where the referendum appears on the ballot.[8]

Due to the complexity of the environment and the many aspects of the referendum that affect who votes and how they vote, the taxing authority cannot perfectly predict how referendum characteristics map into election results. This inability prevents the agenda setter from "manipulating" the referendum to "precisely" sort elections around the majority vote threshold. The inability to perfectly predict is why we would not expect to estimate discontinuities in referendum characteristics when the bandwidth selection is small and only very close elections are used in the

7 See, for example, Ehrenberg et al. (2004), Bowers, Metzger, and Militello (2010), Zimmer et al. (2011), Gong and Rogers (2014), and Dutton (2018).

8 For studies related to election timing, see Berry and Gersen (2011),Anzia (2012), Kogan, Lavertu, and Peskowitz (2018), and Hajnal, Kogan, and Markarian (2022). For research related to ballot placement effects, see Augenblick and Nicholson (2016).

estimation. However, we do expect agenda-setter manipulation to change the composition and distribution of those elections within the relatively wide bandwidths often selected when using local tax referendum elections.[9] Along with Eggers et al. (2015), we expect that by changing the composition and distribution of those elections in the relatively wide bandwidths through agenda-setting manipulation, a discontinuity is more likely to be estimated for the specific referendum characteristics that are able to be manipulated.

The ability to manipulate referenda depends on existing constraints that vary across the types of taxing authorities and types of referenda proposed. This is why we expect, given the relatively wide bandwidths, that discontinuities are more likely to be estimated at the majority vote threshold for referendum characteristics (i) over which the taxing authority has discretion relative to characteristics where constraints restrict the taxing authority's discretion, (ii) where the costs of manipulating are relatively low, and (iii) that the taxing authority can more accurately map to election outcomes.[10] To illustrate, suppose a taxing authority initially considers placing a referendum on a ballot that specifies a 1 millage rate property tax. Given this millage rate, assume the taxing authority perceives that the voters will comfortably pass the referendum. In this case, a taxing authority that benefits from greater tax revenue may have incentive to reconsider and increase the proposed millage rate from 1 to 1.5 because the benefit associated with an increase in tax revenue if the referendum passes is greater than the cost associated with the reduction in the probability the referendum passes. This cost may be relatively minimal because, although increasing the millage rate likely reduces voter support and makes the election closer, it may not significantly change the probability of passage, only whether the winning percentage is somewhat large or slightly smaller.[11] However, it could change whether the election outcome is within the selected bandwidths and, therefore, used in the estimation of the discontinuity.

To further illustrate, consider a city council interested in placing on a ballot a referendum that involves current general expenditures. Suppose the city council is uncertain how many voters would support this referendum but believe it would be a somewhat close election. The city council also believes that stipulating the tax revenue to be spent on public safety (i.e., police officers and firefighters) would slightly increase the proportion of voters who vote in favor of the referendum. This expected benefit associated with increasing the probability of the referendum passing, which is clearly a function of the ability of the city council to predict how the election outcome varies with the referendum specifics, is not without costs. By stipulating public

[9] As mentioned in the introduction, the optimal bandwidth for our estimates, based on Calonico et al. (2017), ranges from 0.10 to 0.23. The estimates of the discontinuities do not change appreciably when we reduce the bandwidth to 0.05.

[10] The expected cost incurred from a failed referendum vote will also vary across referenda, especially based on how easily a failed referendum can be placed on the ballot in a subsequent election and whether failure results in a new tax not being collected or an existing tax not being renewed.

[11] A similar logic applies to a taxing authority having incentive to decrease the tax rate if a referendum is expected not to pass by a significant number of votes.

safety, the ability of the city council to increase current general expenditures by passing the referendum depends on the fungibility of the tax revenue. Besides fungibility, the cost of stipulating public safety would also likely depend on whether the referendum involved a new tax, because it may be more costly for the referendum to fail if it involved a tax renewal. For a referendum involving a new tax, it may also be less costly to time when the referendum appears on the ballot to coordinate it with other issues on the ballot of importance to those individuals who are likely to vote for the referendum. The ability of the city council to predict how the referendum specifics map into election outcomes would also likely differ between a new tax and a tax renewal and whether the referendum had failed in a prior election. When deciding whether to stipulate the tax revenue to be spent on public safety, the city council will weigh all of these expected benefits and costs. Because this expected benefit depends on the change in the probability of the referendum passing, the city council would have little incentive to stipulate public safety if the referendum would clearly fail or clearly pass irrespective of this decision. The city council would have more incentive to stipulate public safety if their expectation is that the referendum would likely fail without the stipulation and likely pass with the stipulation. For these referenda, the election outcomes are likely within the relatively wide bandwidths recommended by Calonico et al. (2017) and Imbens and Kalyanaraman (2012), and this is why discontinuities at the majority vote threshold are more likely to be estimated on referendum characteristics over which the taxing authority has discretion.

The constraints imposed by the State of Ohio on referendum structure (e.g., tax rate, tax purpose) and timing (i.e., in which election the referendum appears) vary across taxing authorities and types of referenda. For example, State of Ohio regulations restrict county commissioners from proposing a referendum stipulating a property tax rate exceeding 1 mill for an armed forces memorial, 1 mill for hospital facility operations, 0.65 mill for tuberculosis centers, 3 mills for managing forest land, 2 mills for detention facilities, and 0.3 mill for water treatment. Similar state restrictions apply for municipality and township tax rates, along with often more stringent restrictions stipulated in a municipality's or township's charter. The State of Ohio also stipulates not only the type of election (general, primary, or special election) in which the referendum can appear but also the process by which a referendum appears on a ballot. This process, which often varies across jurisdiction and referendum types, can significantly restrict the timing of the referendum. Along with the state restrictions, there are process and timing restrictions imposed by municipality charters. Thus, the number of state restrictions on tax rate and timing of elections for school district referenda are fewer than for municipality, township, and county referenda.[12] Aspects such as voter turnout, age, and political affiliation may capture the heterogeneity in voter preferences that the taxing authority is likely to consider when deciding on

[12] Chapter 5705, Tax Levy Law, of the Ohio Revised Code and Appendix A of the Ohio Secretary of State (2023) publication *Ohio Ballot Questions and Issues Handbook* identify other tax rate restrictions for counties, municipalities, and townships.

which ballot to place the referendum. Thus, we should expect discontinuities in these variables for taxing authorities that have fewer restrictions on the timing of the election.

While all jurisdiction types have some discretion in terms of the stipulated purpose for the tax revenue (e.g., public safety, school buildings, current expenditures, roads), the fungibility of the city's/township's general-fund account may make specifying a certain purpose for the tax less costly, because targeted tax revenue may not change actual expenditures. This fungibility would provide city councils and township trustees more incentive to manipulate the closeness cutoff by selecting the purpose based on the likelihood of the referendum passing, leading to discontinuities in these tax purpose variables at the cutoff. In summary, while these constraints could differentially limit the taxing authority's ability to manipulate election closeness, they could also result in these authorities choosing different dimensions by which to accomplish this manipulation.

III. DATA AND DESCRIPTIVE STATISTICS

A. Data Sources

Using election results from local tax referenda that occurred in Ohio between 2004 and 2012 and details on more than 1 million home sales, this paper provides insight on the validity of using an RD design to estimate the effect of tax referendum passage on outcomes such as home prices.[13] We obtain tax referenda data from the Ohio Secretary of State's website and county boards of elections. We obtain the home sales information from individual county auditors in 63 of the 88 Ohio counties. We gathered data on 16,343 local referendum elections taking place between February 2000 and November 2013. We focus on municipality/township, school district, and county tax referenda and restrict our analysis to referenda with complete information that took place between 2004 and 2012. Our referenda data consist of 8,751 municipality/township, 5,008 school district, and 760 county referenda. Along with the votes for and against the referenda, the Ohio Secretary of State and county boards of elections data include the ballot referendum description. This description allows us to identify the type of tax (i.e., property, sales, or income tax), whether it renewed an existing tax, the tax rate, and how the tax proceeds were to be spent (i.e., purpose of referendum). It also allows us to identify "refloated" referenda: those that were preceded within a year by a similar referendum that failed in the jurisdiction. We use the number of votes for and against the referenda to generate

[13] We focus solely on referenda involving the levy of taxes because Ohio law standardizes the proposal process for these referenda and specifies a particular "taxing authority" that may propose referenda. Nonfiscal referenda do not have a standardized proposal process and may be initiated by a legislative authority or through a petition process.

our running variable, *closeness*, which is defined as (yes votes − no votes)/(yes votes + no votes) for a given referendum.

The Board of Elections at the Ohio Secretary of State office provided individual-level information on all registered voters and listed the specific elections in which each registered voter cast a ballot.[14] In addition, these registered voter files included date of birth and potential party affiliation of each voter, as well as the county, township, municipality, and school district of residence. We construct turnout (defined as the percentage of registered voters who voted for or against the referendum), average voter age, and percentages of voters who are affiliated with the Democratic or Republican parties for each referendum.[15] The Ohio Department of Taxation provides all property tax rates levied by jurisdiction as well as municipality/township revenues and expenditures. The National Center for Education Statistics Common Core of Data provides demographic, student performance, and financial information for school districts. Finally, the United States Census provides demographic and commerce information for counties.[16]

In terms of the home sales, we restrict the sample to single-family homes sold for between \$10,000 and \$2,000,000 with complete housing characteristics. Restricting the sample in this manner results in 1,078,916 single-family home sales. The Appendix (available online) presents summary statistics of the observable home sale characteristics that include sales price, year home was built, number of rooms, number of bedrooms, and square footage.

B. Descriptive Statistics

Table 1 presents the means and standard deviations of the election and referenda characteristics. These summary statistics are provided for those that passed and failed, broken down by each type of jurisdiction: municipalities/townships, school districts, and counties.[17] Table 1 indicates that while 80 percent of municipality/township referenda and 74 percent of county referenda pass, school district referenda have a pass rate of only 55 percent. There are also differences in the number of voters, with an average of 1,976 voters for municipalities/townships, 5,572 for school districts, and 37,154 for counties. For municipalities/townships and school districts, the average number of voters is similar whether the referendum passed or

[14] The Ohio Board of Elections collected individual registered voter information for each general and primary election from 2004 to 2012 and for special elections occurring in 2005 and 2006.

[15] Because Ohio does not allow voters to register with a particular political party, this information on political party affiliation is defined as the political party the voter voted for in the most recent primary election.

[16] Because the Census does not collect information annually for many of the county-level variables, yearly data were constructed through interpolation and extrapolation.

[17] Ohio law allows for the formation of various special-purpose districts with limited authority to tax or borrow. In our analysis, we do not include referenda involving a special-purpose district whose boundaries are different than the township or county.

Table 1

Summary Statistics for Referendum and Election Observables

	Municipalities/Townships			School Districts			Counties		
	All	Passed	Defeated	All	Passed	Defeated	All	Passed	Defeated
Referenda passed	0.80			0.55			0.74		
	(0.40)			(0.50)			(0.44)		
Yes votes	1,185	1,275	820	2,855	3,299	2,319	22,000	26,065	10,385
	(9,068)	(8,961)	(1,912)	(5,219)	(6,253)	(3,526)	(41,258)	(46,387)	(15,453)
No votes	791	706	1,132	2,717	2,406	3,092	15,154	15,501	14,161
	(3,979)	(4,282)	(2,368)	(4,381)	(4,434)	(4,286)	(24,756)	(26,350)	(19,522)
Closeness	0.20	0.30	−0.17	0.02	0.18	−0.17	0.13	0.23	−0.16
	(0.25)	(0.16)	(0.14)	(0.22)	(0.13)	(0.14)	(0.22)	(0.42)	(0.14)
Registered voters	103,965	107,133	91,126	13,465	13,706	13,138	99,763	111,274	66,313
	(671,855)	(683,296)	(623,423)	(21,937)	(22,979)	(20,436)	(179,441)	(199,714)	(91,890)
Turnout	0.34	0.34	0.32	0.44	0.43	0.45	0.40	0.40	0.39
	(0.21)	(0.21)	(0.21)	(0.20)	(0.20)	(0.19)	(0.15)	(0.16)	(0.13)
Average age of voter	53.38	53.40	53.33	54.05	54.12	53.94	54.14	54.09	54.28
	(3.54)	(3.51)	(3.67)	(4.35)	(4.57)	(4.02)	(3.19)	(3.26)	(2.98)
Fraction Democrat-affiliated	0.16	0.17	0.16	0.18	0.18	0.17	0.18	0.18	0.17
	(0.11)	(0.11)	(0.10)	(0.11)	(0.11)	(0.10)	(0.10)	(0.10)	(0.09)
Fraction Republican-affiliated	0.34	0.34	0.34	0.30	0.31	0.30	0.33	0.33	0.33
	(0.12)	(0.12)	(0.13)	(0.13)	(0.14)	(0.12)	(0.12)	(0.12)	(0.11)
Timing, type, and rate									
Tax renewal	0.74	0.86	0.25	0.40	0.61	0.13	0.68	0.83	0.26
	(0.44)	(0.35)	(0.43)	(0.49)	(0.49)	(0.34)	(0.47)	(0.37)	(0.44)
Failed referendum prior year	0.08	0.05	0.23	0.29	0.21	0.38	0.14	0.10	0.23
	(0.28)	(0.21)	(0.42)	(0.45)	(0.41)	(0.49)	(0.34)	(0.30)	(0.42)

Proposed property tax	0.94	0.97	0.83	0.73	0.79	0.65	0.91	0.94	0.82
	(0.24)	(0.18)	(0.37)	(0.45)	(0.40)	(0.48)	(0.29)	(0.24)	(0.38)
Income tax rate	0.63	0.62	0.64	0.81	0.81	0.81			
	(0.51)	(0.51)	(0.51)	(0.30)	(0.25)	(0.33)			
Millage rate	1.99	1.93	2.29	5.37	4.83	6.16	1.11	1.15	0.98
	(1.51)	(1.44)	(1.75)	(2.98)	(3.02)	(2.74)	(0.93)	(0.96)	(0.78)
Sales tax rate							0.46	0.45	0.46
							(0.20)	(0.21)	(0.19)
Purpose									
Public safety	0.46	0.49	0.31				0.15	0.15	0.17
	(0.50)	(0.50)	(0.46)				(0.36)	(0.35)	(0.38)
Senior/Community center							0.19	0.23	0.08
							(0.40)	(0.42)	(0.27)
Health							0.33	0.34	0.32
							(0.47)	(0.47)	(0.47)
Current expenditures	0.20	0.20	0.22	0.35	0.33	0.37			
	(0.40)	(0.40)	(0.41)	(0.48)	(0.47)	(0.48)			
Roads	0.17	0.16	0.22						
	(0.37)	(0.36)	(0.42)						
Emergency expenditures				0.26	0.26	0.26			
				(0.44)	(0.44)	(0.44)			
Capital expenditures				0.21	0.24	0.18			
				(0.41)	(0.43)	(0.38)			
Observations	8,751	7,021	1,739	5,008	2,738	2,270	760	563	197

Note: Of the 8,751 municipality/township observations, 6,559 have registered voter and party affiliations information. Of the 5,008 school district observations, 3,725 have registered voter and party affiliations information, 4,790 have tax rate information, and 4,502 have tax base information. Of the 760 county observations, all have registered voter information and 746 have party affiliations information. Standard deviations are in parentheses.

failed. However, county referenda that passed have more votes cast than county referenda that fail. While the number of registered voters averages 13,465 for school district referenda and around 100,000 for the other jurisdictions, turnout is slightly higher for school district referenda (44 percent) and slightly lower for municipalities/townships (34 percent).

Table 1 also indicates that voters' party affiliation and average age do not vary appreciably across jurisdiction type or referendum outcome. For all three types of jurisdictions, the fraction affiliated with the Democratic Party is slightly less than 20 percent, and around a third of voters are affiliated with the Republican Party. Average voter age is slightly above 50 years for all jurisdiction types. The fraction of tax renewal referenda is much larger for municipalities/townships and counties compared with school districts, and the probability that a tax renewal referendum passes is also much higher than that of a nonrenewal referendum for all three types of jurisdictions.[18] Finally, in terms of the referendum elections, Table 1 indicates that the probability a referendum was preceded within a year by a similar referendum that failed is much higher for school districts and much lower for municipalities/townships. These similar referenda that were placed on a subsequent ballot had a high probability of failing again.

Besides differences in the election characteristics across jurisdiction types and referendum passage, Table 1 also indicates variation in the type of tax and the tax rate stipulated. While Ohio allows municipalities/townships and school districts to levy property and income taxes, counties are restricted to levying either a property or sales tax. More than 90 percent of the municipality/township and county referenda involve a property tax, and 73 percent of school district referenda stipulated a property tax. For all three types of jurisdictions, the probability that a property tax referendum passes is greater than that of the alternative tax available to the jurisdiction. For municipalities/townships and school districts, property tax referenda stipulating a relatively high millage rate were more likely to fail than referenda with a lower millage rate, and school district millage rates are much greater than for the other types of jurisdictions. In addition to varying across the type of tax and tax rates, the probability of passage also varies across the type of expenditure the referendum stipulates. Table 1 indicates that referenda pertaining to public safety are much more likely to pass than other municipality/township referenda, referenda involving capital expenditures are more likely to pass than other school district referenda,[19]

[18] We define a tax renewal as (i) when an existing tax is set to expire and passing the referendum would extend the tax at the rate for a longer period of time or (ii) when a passed referendum extends, increases, or decreases a tax currently in place and serves the same purpose as the tax levy it supersedes. The majority of these tax renewals occur when an existing tax is set to expire, and there are a few cases where jurisdictions have bundled a renewal and a new tax in one referendum.

[19] The higher pass rate of referenda involving capital expenditures is likely attributable to the Ohio School Facilities Commission, which provides large state subsidies to many school districts interested in improving their capital stocks. Conlin and Thompson (2017) provide a detailed description and analysis of this State of Ohio program.

and referenda involving senior/community centers are more likely to pass than other county referenda.[20]

IV. VALIDITY OF RD IDENTIFICATION STRATEGY

While the descriptive statistics indicate that the probability of a referendum passing depends on election and referendum characteristics, they do not address the validity of the RD design. An assumption when using an RD design to infer the causal effect of referenda passage on the outcome of interest is that the referendum cannot be manipulated to change the composition and distribution of elections in the selected bandwidth so as to affect the estimated discontinuity at the cutoff. While in most cases it is difficult to directly test this assumption, plotting the histogram or kernel density of the observations across the closeness measure and observing whether or not a discontinuity in the density occurs at the cutoff can provide suggestive evidence as to the validity of this assumption.

Panel A of Figure 1 presents kernel densities of referenda across the closeness measure, separately for municipalities/townships, school districts, and counties. While these figure panels show that the central tendency of these densities differs, reflecting the differing mean passage rates across jurisdiction described in Subsection III.B, here we are most interested in what is happening to the density at the closeness cutoff (i.e., where *closeness* $= 0$). To provide a statistical test for whether the density is continuous at the cutoff, we use the McCrary (2008) density test, which is standard in RD designs. The test indicates a decrease in probability across the closeness threshold for each of these three distributions, but none of these decreases are statistically significant. The first graph in Panel B of Figure 1 presents the kernel density for the more than 14,000 referenda from all three types of jurisdictions, and the discontinuity continues to be statistically insignificant. The lack of statistical significance is not surprising given that the taxing authority may have minimal discretion in terms of the structure and timing for a subset of these referenda.[21] In particular, we expect the taxing authority to have less discretion for referenda involving a tax renewal or involving

[20] The expenditures classifications are for all those that make up more than 10 percent of the total number of referenda. The classifications for municipality/township referenda are public safety, capital, senior/community center, current expenditures, general, health, roads, library, parks/recreation, sewer, and flood. The expenditure classifications for school district referenda are library, avoid deficit, capital, current expenditures, emergency, general bond, and recreation. The expenditure classifications for county referenda are public safety, capital, senior/community center, current expenditures, general, health, roads, library, parks/recreation, community college, zoo, and child services.

[21] This lack of statistical significance is consistent with much of the previous literature that uses an RD design with the running variable being closeness, with Hong and Zimmer (2016) and Dronyk-Trosper (2017) being notable exceptions when focusing on specific types of elections. When considering Michigan school bond elections and Massachusetts education overrides, Hong and Zimmer (2016) and Dronyk-Trosper (2017) only find a discontinuity in the probability across the closeness cutoff when restricting the data set to referenda that appear on multiple ballots and debt exclusion referenda, respectively.

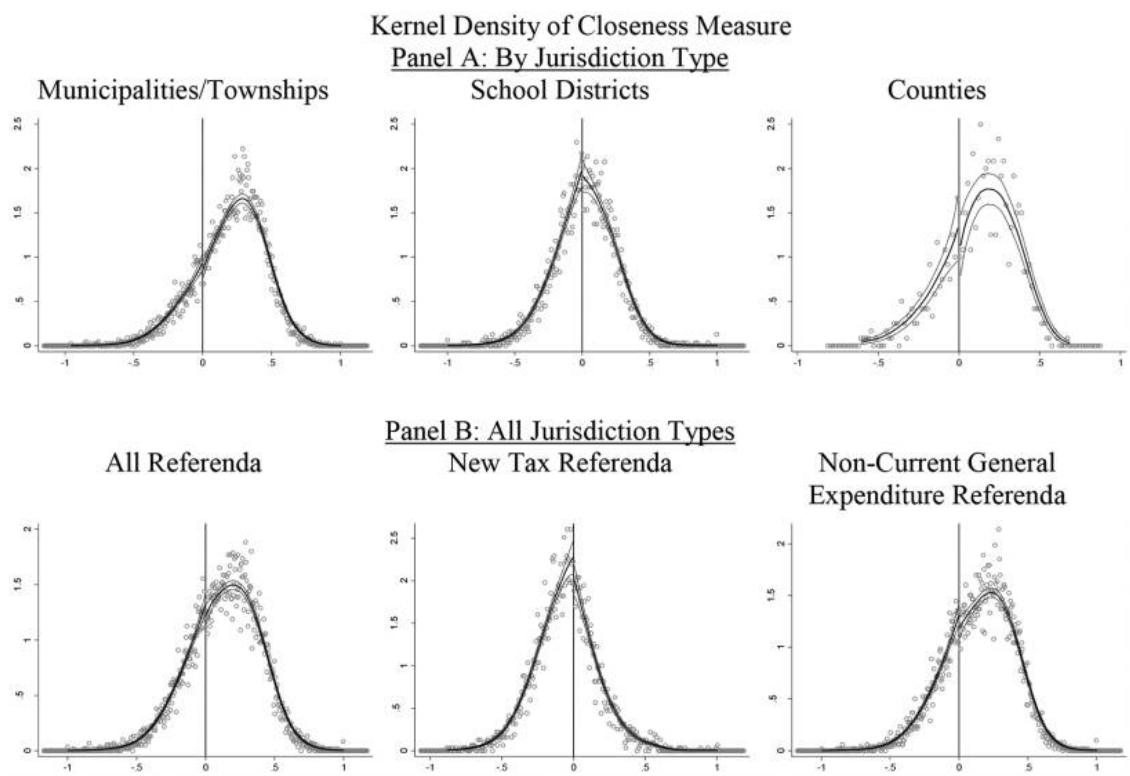

Figure 1. Kernel density of closeness measure. Panel A: By jurisdiction type. Panel B: All jurisdiction types. A color version of this figure is available online.

current general expenditures. The other two graphs in Panel B exclude these types of referenda and depict the distributions for the 5,546 referenda that pertain to a new tax and the 10,989 referenda that do not pertain to current general expenditures. When including only these referendum types, the discontinuity at the closeness cutoff increases and becomes statistically significant at the 10 percent level. In both graphs, however, there is a decrease in probability across the closeness cutoff. One possible explanation for this decrease is that referenda that fail by a small margin are often modified and placed on a subsequent ballot.

The next step in assessing the validity of the RD design is to conduct the covariate balance tests and assess whether there is an estimated discontinuity at the cutoff for characteristics of the referendum and election that are able to be manipulated. We present separate tests for the three types of jurisdictions because the different taxing authorities have varying constraints based on the type of referendum. The referendum and election characteristics on which we conduct covariate balance tests include the type of tax proposed (property, income, or sales), the tax rate, the purpose of the tax revenue, voter turnout, average voter age, voter political party affiliation, whether the referendum involved a new tax or a tax renewal, and whether a similar referendum had been defeated in the jurisdiction within the past year. Despite the potential for manipulation of some of these variables as described in Section II, few if any of the previous studies using this tax election–based RD design examine these dimensions when conducting their covariate balance tests, leaving an open question around whether these factors are problematic for the validity of the empirical design.[22]

We estimate discontinuities in the referendum and election characteristics imposing the mean squared error-optimal (MSE-optimal) bandwidth restrictions proposed by Calonico et al. (2017). Given these bandwidth restrictions, we estimate the RD specification:

$$\begin{aligned} y_{re} = {} & \alpha + \beta_1 1[closeness_r \geq 0] + \beta_2(closeness_r \times 1[closeness_r \geq 0]) \\ & + \beta_3(closeness_r \times 1[closeness_r < 0]) + \varepsilon_{re}, \end{aligned} \tag{1}$$

where y_{re} is the referendum or election characteristic; $closeness_r$ is (yes votes − no votes)/(yes votes + no votes) for referendum r; $1[closeness_r \geq 0]$ and $1[closeness_r \leq 0]$ are indicator variables for the referendum r passing and failing, respectively; and ε_{re} is an idiosyncratic error term. The estimate of β_1 quantifies the discontinuity in referendum or election characteristic y_{re} at the closeness cutoff. This specification is estimated separately for the municipality/township, school district, and

[22] Much of the previous literature (mostly related to school district referenda) considers underlying community demographic variables, as well as baseline spending and revenue characteristics, of these communities in these covariate balance tests. While these may attempt to uncover discontinuities in underlying community preferences, they are unlikely to be sufficient for capturing information about the underlying agenda-setting behavior we describe in this paper.

county referenda and uses only those elections that satisfy the MSE-optimal bandwidth constraint.

Table 2 provides the estimates associated with the covariate balance tests for the referendum and election characteristics. The first column for each jurisdiction type provides the estimate of the discontinuity (i.e., β_1) when the bandwidths above and below the closeness cutoff are restricted to be the same, and the second column contains the estimates when these bandwidths are allowed to differ. As indicated in the Appendix, the optimal bandwidths vary based on characteristic and jurisdiction type. These bandwidths range from 0.10 to 0.23 when restricted to be the same above and below the cutoff and from 0.10 to 0.31 when bandwidths are allowed to differ.[23]

Because of the discretion school boards have over tax rates, the discretion both school boards and county commissioners have over when the referendum appears on the ballot, and the fungibility of municipality and township general-fund expenditures, we expect discontinuities are more likely for tax rates in school district referenda, voter characteristics in school board and county referenda, and purpose of the tax revenue for municipality/township referenda. The coefficient estimates in Table 2 corresponding to these factors are in bold. We expect the coefficient estimates that are not in bold to be economically and statistically insignificant because they correspond to discontinuities in aspects of the referendum or election that the taxing authority has less ability to manipulate due to constraints imposed by the state and local jurisdictions.

The estimates in Table 2 indicate that all jurisdiction types have only a minimal discontinuity associated with referenda involving a proposed property tax, and only school district referenda have a statistically significant discontinuity in proposed tax rates at the closeness cutoff. Irrespective of whether the bandwidth is restricted to be the same above and below the cutoff, school district referenda that barely pass have income tax rates that are 0.18 percentage points higher. The discontinuities in school district income tax and millage rate increase when we include only referenda proposing a new tax, consistent with taxing authorities having fewer constraints on their agenda-setting abilities for new taxes.[24] The estimates associated with tax rates are consistent with school boards having fewer constraints on the tax rate, especially for referenda involving a new tax, than the other taxing authorities. In terms of the purpose stipulated for the revenue, the only economically and statistically significant

[23] The Appendix presents the graphical depictions of how the election and referendum characteristics vary based on closeness. The Appendix also contains estimated discontinuities at the closeness cutoff for a variety of bandwidths including those proposed by Calonico, Cattaneo, and Titiunik (2014), Cattaneo, Idrobo, and Titiunik (2019), and Imbens and Kalyanaraman (2012). The Appendix also contains estimates for a variety of bandwidths that range from 0.05 to 0.25. As mentioned in footnote 6, the standard errors do increase as the bandwidth becomes smaller, but the point estimates do not change appreciably from those in Table 2.

[24] The Appendix contains analogous estimates to those in Table 2 when renewal tax referenda and current general expenditures are excluded from the estimation.

Table 2

Discontinuity Coefficients on Referenda and Election Characteristics

	Municipalities/Townships		School Districts		Counties	
	MSE-Optimal		MSE-Optimal		MSE-Optimal	
	Same Bandwidths	Different Bandwidths	Same Bandwidths	Different Bandwidths	Same Bandwidths	Different Bandwidths
Type and rate						
Proposed property tax	−0.013 (0.027)	−0.011 (0.026)	**−0.007 (0.035)**	**0.003 (0.037)**	−0.174 (0.107)	−0.137 (0.094)
Income tax rate	0.223 (0.183)	0.228 (0.185)	**0.183** (0.089)**	**0.179** (0.088)**		
Millage rate	0.035 (0.150)	0.023 (0.137)	**−0.272 (0.267)**	**−0.203 (0.263)**	0.234 (0.261)	0.440 (0.220)
Sales tax rate					−0.386 (0.287)	−0.388 (0.275)
Public safety	**0.012 (0.042)**	**012 (0.044)**			0.065 (0.105)	0.069 (0.101)
Senior/Community center					−0.133 (0.091)	−0.189 (0.095)
Health					0.043 (0.097)	0.051 (0.101)
Current expenditures	**0.050 (0.035)**	**0.062 (0.035)**	−0.033 (0.045)	−0.020 (0.041)		
Roads	**−0.066* (0.037)**	**−0.065* (0.036)**				
Emergency expenditures			0.006 (0.038)	0.009 (0.037)		
Capital expenditures			0.041 (0.036)	0.031 (0.033)		
Election timing						
Turnout	0.023 (0.026)	0.022 (0.025)	**0.029 (0.019)**	**0.031* (0.018)**	**0.039 (0.040)**	**0.049 (0.040)**
Average age of voter	−0.158 (0.398)	−0.093 (0.385)	**−0.191 (0.418)**	**−0.181 (0.409)**	**−1.72** (0.796)**	**−1.72** (0.819)**
Fraction Democrat-affiliated	0.004 (0.011)	0.003 (0.011)	**0.004 (0.012)**	**0.004 (0.012)**	**−0.018 (0.018)**	**−0.023 (0.018)**
Fraction Republican-affiliated	0.005 (0.012)	0.007 (0.012)	**0.001 (0.012)**	**0.0002 (0.011)**	**−0.024 (0.037)**	**−0.036 (0.036)**
Tax renewal	0.034 (0.047)	0.030 (0.045)	0.022 (0.040)	0.028 (0.039)	−0.022 (0.115)	−0.040 (0.126)
Failed referendum prior year	0.018 (0.032)	0.018 (0.030)	0.003 (0.043)	0.003 (0.042)	−0.074 (0.093)	−0.059 (0.090)

Note: The first and second columns for each jurisdiction type provide the estimate of the discontinuity when a linear function of closeness is fitted to referenda for the MSE-optimal bandwidth when the bandwidths above and below the cutoff are restricted to be the same and are allowed to differ, respectively. (See Calonico et al. (2017) for an overview of how these bandwidths are derived.) As indicated in the Appendix, the optimal bandwidths vary based on characteristic and jurisdiction type. These bandwidths range from 0.10 to 0.23 when restricted to be the same above and below the cutoff and from 0.10 to 0.31 when bandwidths are allowed to differ. Clustered standard errors at the jurisdiction level are in parentheses.

* $p < 0.1$.

** $p < 0.05$.

discontinuities are for municipality/township referenda pertaining to roads. These estimates suggest that municipality/township referenda that barely pass are approximately 6.6 percentage points less likely to be for roads than those that barely fail. Similar to the school district tax rate discontinuities, the estimated discontinuities for municipality/township road referenda increase when only referenda involving a new tax are used in the estimation. We thought discontinuities on the purpose of the tax revenue might exist because resources for these different purposes are often obtained from the city's/township's general-fund account and likely fungible. Therefore, if a city passes a referendum that generates a million dollars in revenue, that does not mean the amount spent on police or road repair would vary depending on whether the referendum stipulated public safety or roads. This fungibility, which is likely less available to school boards and county commissioners, provides city councils and township trustees more incentive to influence the election outcome through their selection of the tax purpose.

Along with the structure of the referendum, taxing authorities may manipulate closeness through their decision regarding when to place the referendum on the ballot. The estimates in Table 2 suggest that voter turnout and average age of voter differ across the closeness cutoff for school district and county referenda, respectively. While the discontinuities in our voter-composition measures at the cutoff are minimal for school districts, school district referenda that barely pass have turnout that is 3 percentage points higher than those that barely fail. The estimated discontinuity in voter turnout is slightly larger for counties than for school districts, but those associated with school district referenda are statistically significant when bandwidths are allowed to differ. There are economically and statistically significant discontinuities in voter age composition for county referenda. They suggest that county referenda that barely pass have voters who are significantly younger than referenda that barely fail.[25] School boards and county commissioners can influence voter turnout and voter composition and affect the probability a referendum passes, by selecting an election when individuals who favor the referendum are more likely to vote. These preferences are likely correlated with an individual's age for not only county and school district referenda but also municipality/township referenda. However, city councils and township boards often have less discretion over election timing, which explains why the estimates for municipality/township referenda suggest no discontinuity in turnout, voter age, or voter political affiliation.

The estimates in Table 2 use all of the referenda for the three different jurisdiction types. While we expect taxing authorities to have less discretion to manipulate a referendum involving a tax renewal, it may be the case that the taxing authority can make a more accurate prediction of how the election outcome will vary with the

[25] As indicated in the Appendix, when only referenda involving a new tax are used in the estimation, these estimated discontinuities associated with voter turnout and voter age increase significantly for school district and county referenda, and the discontinuities associated with fraction democratic affiliated become statistically significant for county referenda.

referendum structure for those involving a tax renewal or a referendum that previously failed. However, the estimates in Table 2 provide minimal evidence of a discontinuity at the closeness cutoff for both of these types of referenda.

Because the probability of referendum passage is also correlated with jurisdiction characteristics that are not related to the referendum,[26] we also conduct covariate balance tests for a large set of jurisdiction characteristics that the taxing authorities are unlikely able to manipulate (e.g., population, previous governmental expenditures). As indicated in the appendix tables, almost all of the discontinuity estimates for these 41 jurisdiction characteristics are close to zero, and none are statistically significant. This lack of discontinuities at the closeness cutoff is consistent with the existing literature and is used to support the validity of these RD designs.[27] However, as our other results suggest, the estimated discontinuities in the election and referendum characteristics that officials have discretion over — characteristics that are less often cited in the existing literature — pose potential problems for the validity of these designs.[28]

There are several key takeaways from these covariate balance tests. First, several election and referendum characteristics have noticeable estimated discontinuities, including the purpose of the tax for municipality/township referenda, the tax rate and voter turnout for school district referenda, and average age of voter for county referenda. As noted in Section II, these differences are likely due to the discretion the taxing authorities have in terms of referendum structure or timing to manipulate election closeness. Specifically, city councils and township trustees do not find it costly to restrict how the tax revenue associated with referendum passage will be spent (due to fungibility), school boards have minimal binding constraints in terms of the tax rate and when the referendum appears on the ballot, and county commissioners have minimal constraints in terms of when the referendum appears on the ballot. Second, for almost all of the characteristics corresponding to election

[26] As presented in the Appendix, the probability of passing a referendum is higher in municipalities/townships with larger revenues and expenditures. The distribution of expenditures also varies with election outcome, with municipalities/townships more likely to pass referenda involving capital expenditures and less likely to pass fire/police and debt referenda. These findings also indicate that student exam performance is better for school districts that pass referenda, and demographic characteristics were similar for districts where the referendum passed or failed. Finally, these results indicate that counties that pass a referendum are larger in terms of population, number of households, and commercial activity than counties where the referendum was defeated.

[27] Previous literature primarily focuses on demographics, enrollment, teacher characteristics, revenues, and expenditures when conducting these covariate balance tests on school district referenda.

[28] Cellini (2009) includes the size of the bond in these tests, while Hong and Zimmer (2016) and Isen (2014) are the only studies that consider bond and tax type characteristics. None of the previous studies using bond and tax election results consider characteristics of the people who turn out to vote in these elections, despite the fact that extensive research has documented how these voter characteristics play a role in explaining election outcomes (Button, 1992; Ehrenberg et al., 2004; Gradstein and Kaganovich, 2004; Berkman and Plutzer, 2004; Fletcher and Kenny, 2008; Shober, 2011; Zimmer et al., 2011; Bowers and Lee, 2013).

and referendum aspects over which the taxing authority has less discretion, the estimated discontinuities are close to zero and not statistically significant.

V. IMPLICATIONS FOR HOME SALES ANALYSES

Much of the economics literature using these types of RD designs attempts to examine the causal impact of revenue changes on outcomes like housing prices. We conduct a similar analysis here to assess the implications that the discontinuities we estimated in Section IV have for this type of home sales analysis. To account for the varying composition of homes being sold, we regress ln(sales price) on home characteristics to obtain the residuals. For the more than 1 million sales that occurred within a year after a referendum election, we use these residuals as the dependent variable to estimate the discontinuity at the closeness cutoff using the specification in Equation (1). The estimates of the discontinuities from this specification are contained in Panel A of Table 3 for MSE-optimal bandwidths with and without the restriction that the bandwidths are the same above and below the cutoff.[29] Focusing on the estimates when the bandwidths are restricted to be the same, the point estimates in Panel A of Table 3 suggest a 1.04 percent increase and a 0.48 percent decrease in housing prices across the closeness cutoff for municipalities/townships referenda and school districts referenda, respectively.[30] Numerous papers using election closeness as the running variable in an RD identification strategy interpret these types of estimates in a causal manner.

Given that our covariate balance tests indicated numerous manipulatable referendum/election characteristics that vary across the cutoff, we are concerned that the discontinuity estimates in Panel A suffer from omitted-variable bias. Our concern associated with omitted-variable bias is based on the fact that the referendum/election characteristics are capitalized in, and therefore likely correlated with, home prices. For example, home prices are expected to be lower in jurisdictions with higher tax burdens compared with lower tax burdens for similar levels of public goods and services. With the same tax burden, we would also expect higher home prices in jurisdictions that are safer due to a greater police presence and those with a community center and better roads. Furthermore, higher tax rates are likely associated with higher levels of expected growth in these communities, which would also be correlated with home prices.

[29] Note that when allowed to differ, the bandwidth below the closeness cutoff is significantly smaller than the bandwidth above the cutoff. Because of this, we focus on the estimates where the bandwidths are restricted to be the same above and below the cutoff. Unlike the estimates from the covariate balance tests, these estimates using home sale residuals are sensitive not only to the restriction of the bandwidths being the same but also to how the bandwidths are selected. See tables in the Appendix.

[30] The point estimates of the discontinuities for municipalities/townships and school districts referenda are more realistic than those for county referenda.

Table 3

Discontinuity Coefficients on Home Sales Residuals

	Home Sales within a Year after Referendum Election					
	Municipality/Township		School Districts		Counties	
	MSE-Optimal		MSE-Optimal		MSE-Optimal	
	Same Bandwidths	Different Bandwidths	Same Bandwidths	Different Bandwidths	Same Bandwidths	Different Bandwidths
Panel A: No referendum/election covariates						
Discontinuity (β_1)	0.0104***	0.0062***	−0.0048	−0.0638***	−0.1200***	−0.1461***
	(0.0026)	(0.0015)	(0.0030)	(0.0012)	(0.0169)	(0.0248)
Bandwidth	0.12	0.02, 0.23	0.07	0.003, 0.19	0.04	0.02, 0.12
Observations	121,162	154,333	187,690	213,720	32,399	108,107
Panel B: With referendum/election covariates						
Discontinuity (β_1)	0.0065	−0.0329***	0.0845***	−0.0663***	−0.1357**	−0.4222***
	(0.0045)	(0.0020)	(0.0042)	(0.0020)	(0.0630)	(0.0402)
Bandwidth	0.11	0.01, 0.23	0.03	0.003, 0.08	0.01	0.02, 0.02
Observations	72,473	99,989	66,723	81,782	10,886	14,398
Panel C: No referendum/election covariates with Panel B bin size selection						
Discontinuity (β_1)	−0.0093**	−0.0548***	0.0740***	−0.0331***	−0.3082***	−0.2232***
	(0.0042)	(0.0018)	(0.0033)	(0.0016)	(0.0630)	(0.0410)
Bandwidth	0.11	0.01, 0.23	0.03	0.003, 0.08	0.01	0.02, 0.02
Observations	72,543	99,894	66,900	81,787	10,886	14,398

Note: The first and second columns for each jurisdiction type provide the estimate of the discontinuity when a linear function of closeness is fitted to referenda for the MSE-optimal bandwidth when the bandwidths above and below the cutoff are restricted to be the same and are allowed to differ, respectively. These bandwidths and the corresponding number of observations are specified. The bandwidths in Panels B and C are the same to the third digit to the right of the decimal but slightly differ due to rounding error. Clustered standard errors at the municipality and school district level are in parentheses. Due to convergence issues, we are unable to cluster for the county specifications.

** $p < 0.05$.

*** $p < 0.01$.

To provide insight on this possible misspecification, we follow the suggestion of Lee and Lemieux (2010) to see how the discontinuity estimates change when the referendum/election characteristics are included as covariates in the RD specification in Panel A (those noted in Table 2).[31] Panel B of Table 3 reports estimates of the discontinuities when these characteristics are included as covariates and bandwidth selection is based on MSE-optimal bandwidths. First, note that the MSE-optimal bandwidths vary significantly based on whether the covariates are included in the estimation. This makes it difficult to identify whether the differences in the Panel A and Panel B estimates are due to the addition of covariates or due to the change in the set of observations in the bandwidths. For this reason, Panel C presents the discontinuity estimates without the referendum/election characteristics as covariates but using the MSE-optimal bandwidths in Panel B.

As indicated in Table 3, the discontinuity estimates usually change by an order of magnitude when these referendum/election covariates are included. For example, when these covariates are not included, the estimates suggest that municipality/township referenda that barely pass decrease home prices by 0.93 percent compared with those that barely fail, while the estimates when the covariates are included suggest an increase in home prices of 0.65 percent.[32] The fact that the estimates in Panels B and C of Table 3 differ significantly is not surprising given the results of our covariate balance tests and suggests omitted-variable bias in the estimates that do not include these referendum/election covariates in the model.

The implications of the estimates in Table 3 for other studies that use election closeness in an RD design are twofold. First, much of the prior literature does not include any additional covariates beyond the history of prior election outcomes in the RD specification. For studies that use outcomes of elections where referendum/election characteristics are able to be manipulated and the bandwidth selection is relatively wide, our results suggest that failure to control for these characteristics may result in omitted-variable bias in the discontinuity estimate. Second, failure to control for these characteristics is likely more problematic in studies that examine a host of different types of referenda with very heterogeneous purposes, such as both operating levies and capital bond referenda or multiple kinds of operating levies, as is the case in studies such as our own and that of Isen (2014) and Abott et al. (2020).

[31] Lee and Lemieux (2010, p. 335) state "the inclusion of baseline covariates — no matter how highly correlated they are with the outcome — should not affect the estimated discontinuity, if the no-manipulation assumption holds. If the estimates do change in an important way, it may indicate a potential sorting of the assignment variable that may be reflected in a discontinuity in one or more of the baseline covariates." Lee and Lemieux (2010) also indicate that another reason to include these additional covariates is to obtain a more precise estimate of the discontinuity.

[32] The only specification where the discontinuity estimate does not change appreciably is for school district referenda where the bandwidth is restricted to be the same above and below the cutoff. In this case, the estimates in Panels B and C suggest that school district referenda that barely pass increase home prices by 8.45 and 7.40 percentage points more than those that barely fail when referendum/election characteristics are and are not included as covariates, respectively.

Perhaps controlling for these characteristics is less important in studies, such as Cellini, Ferreira, and Rothstein (2010) and Hong and Zimmer (2016), that specifically examine school bond referenda for new construction/renovations, because the purpose of these referenda is more homogenous. That said, even in studies that analyze more homogenous referenda or control for a large set of observable referendum/election characteristics, unobservable differences that influence the ability of local officials to manipulate both election outcomes and housing prices may continue to bias these RD estimates when bandwidth is relatively wide. Even after controlling for numerous characteristics in Panel B of Table 3, we are cautious in interpreting the discontinuity estimates because we are aware of numerous other dimensions that we are unable to control for, most notably the quality of the taxing authority itself. Higher-quality local officials are likely more knowledgeable about their constituents' preferences and therefore better able to engage in the agenda-setting manipulation behavior described in this paper. These higher-quality local officials would also be expected to be operating more high-functioning, efficient governments, which should be positively capitalized into the housing market. In summary, we urge caution in assigning causal language to the results of tax election–based RD designs with relatively wide bandwidths where agenda-setting behavior is possible.

VI. CONCLUSION

There is an extensive literature using RD identification strategies with election closeness as the running variable to empirically address impacts of local funding changes on outcomes, such as housing prices. In the ideal setting, these election-based RD designs overcome the issues associated with adequately controlling for differences in communities that experience changes in their local tax revenue by comparing home prices in communities that just barely pass a tax referendum to those in communities where tax referendum barely fail. In practice, however, limitations arise when the discontinuity estimates are based on election outcomes that are not extremely close to the majority vote threshold, leading to biased estimates of these housing price discontinuities. While local officials are unlikely to precisely manipulate election outcomes at the majority vote threshold, these agenda setters often have discretion over multiple referenda characteristics, and this discretion enables them to manipulate election outcomes within the larger bandwidths we — and other papers using these election-based RD designs — consider.

To assess the validity of the RD design in this context, we examined election results from municipality, township, school district, and county tax referenda in Ohio. Consistent with much of the existing research, we did not find a statistically significant discontinuity in the histogram of referenda outcomes at the closeness cutoff. However, we did find statistically significant discontinuities in this histogram — signifying potential manipulation of the closeness cutoff (and a violation of one of the key assumptions of the RD method) — when we consider only referenda that pertain to a new tax (i.e., not a renewal tax) and only referenda that do not pertain to current general expenditures. These are the referendum types over which we expected taxing authorities to have more discretion in terms of the structure and timing and, thus, greater ability to manipulate the cutoff.

To assess the other key assumption of the RD method — ensuring that observations are comparable on both sides of the cutoff — we used our detailed information on referendum and election characteristics to evaluate covariate balance tests on aspects of the referendum over which taxing authorities have discretion. Unlike existing research, which does not consider these characteristics, we examined whether agenda-setting manipulation results in estimated discontinuities in these covariates when bandwidths are relatively wide. Using Calonico et al. (2017) bandwidths, we estimated economically and statistically significant discontinuities at the closeness cutoff in the covariate balance tests pertaining to referendum and election characteristics over which the taxing authorities have discretion, including (1) tax rates and voter turnout for school district referenda, (2) voter age for county referenda, and (3) the tax revenue purpose for municipality/township referenda. We also did not find statistically significant discontinuities in the covariate balance tests pertaining to characteristics over which taxing authorities have less discretion. The discontinuities in these covariate balance tests create omitted-variable bias in the housing price discontinuity estimates. Specifically, we showed that when including referendum/election covariates, the housing price discontinuity estimates change appreciably, suggesting that failure to control for these characteristics — as much of the previous literature does — may result in biased estimates of the housing price discontinuities.

This research provides guidance to researchers interested in using an RD identification strategy based on election closeness as the running variable. It suggests that researchers should consider the discretion that taxing authorities have in terms of the structure of the referendum and which election the referendum appears on the ballot. The greater the agenda-setting ability, the greater the concern that election closeness may be manipulated in a manner that compromises the validity of the RD design when relatively wide bandwidths are selected. The issues noted in this paper may be most relevant to studies that examine a host of different types of referenda with very heterogeneous purposes and less of an issue where the purpose of the referenda is more homogenous. In all of these studies, however, the possibility of unobservable factors influencing the ability of local officials to manipulate both election outcomes and housing prices remains. Thus, we urge caution in assigning causal language to the results of tax election–based RD designs with relatively wide bandwidths where agenda-setting behavior is possible. Finally, researchers should also consider the incentives facing taxing authorities in these settings. Taking into consideration heterogeneity across taxing authorities in terms of objective functions and agenda-setting ability is important when evaluating the appropriateness of using election closeness as the running variable in an RD estimation procedure. For elections where agenda-setting ability is likely, taking models that endogenize the structure of what is being voted on and the timing of the election to the data seems like the appropriate next step.

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